

Comparison

Black Box vs White Box testing
Functional vs. Non Functional testing
Verification VS Validation (static testing VS dynamic testing)

Black Box vs White Box testing

Black box vs white box

Black Box	White Box
Done by software testers	Done by developers
Do not need to know the internal code	Need to know the internal code and implementation
Verifying the functionality and requirement	Verifying the structure of the technical design document
Do not need to know the programming language	Need to know the programming language
Regressing testing, system testing, acceptance testing	Statement coverage, condition coverage, loop coverage

Functional vs. Non Functional testing

Comparison: Functional vs. Non-Functional Testing

Basis of Comparison	Functional Testing	Non-Functional Testing
Primary Goal	Validates what the system does.	Validates how the system performs.
Focus Area	Focuses on User Requirements and business specifications.	Focuses on User Expectations and quality attributes.
Execution	Can be easily done manually , though automation is also widely used.	Difficult to do manually; almost always requires automation tools .
Requirement Source	Based on Functional Specifications .	Based on Technical Parameters (speed, stability, etc.).
Key Examples	Unit Testing, Integration Testing, System Testing, and Regression Testing.	Load Testing, Stress Testing, Volume Testing, and Security Testing.

Verification VS Validation (static testing VS dynamic testing)

Verification (static testing)	Validation (dynamic testing)
Process of testing documents manually (SRS, Plan, Design, code review)	Testing the product functionality by execution
Does not involve any code execution	Involves code execution to validate the functionality
Find bugs or defects in early stage	Find bugs or defects while the application is running
Cost of defect is low	Cost of defect is high
Reviews, technical reviews, walk through, code review	Unit testing, integration testing, system testing, acceptance testing, regression testing